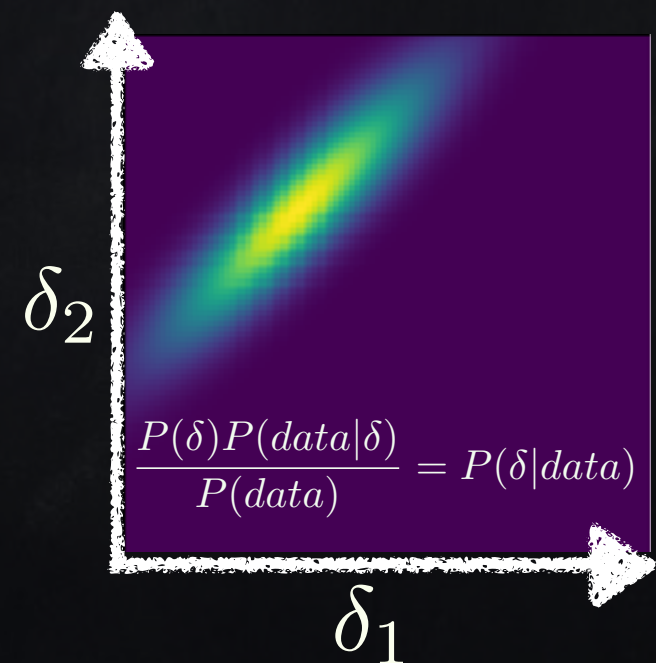




Imperial College
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Bayesian inference methods for ecology

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Bayesian inference methods for ecology

Theory

What is Bayesian inference?

Probability 101

Bayes' theorem in biology

Priors and likelihood

Interpretation of posteriors

Markov Chain Monte Carlo

Applications

Population dynamics in
microscopic and
macroscopic communities

Phylogeny and model
selection

ODE generative models:
Morphogen patterning of
embryonic tissues

Bayesian inference methods for ecology

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At the end of today's
class you should...

- Understand what is Bayesian inference and why is important in Biology
- Be confident with the manipulation of probability distributions
- Identify the basic steps in applying Bayesian inference
- Interpret the biological meaning of a posterior distribution

Bayesian inference methods for ecology

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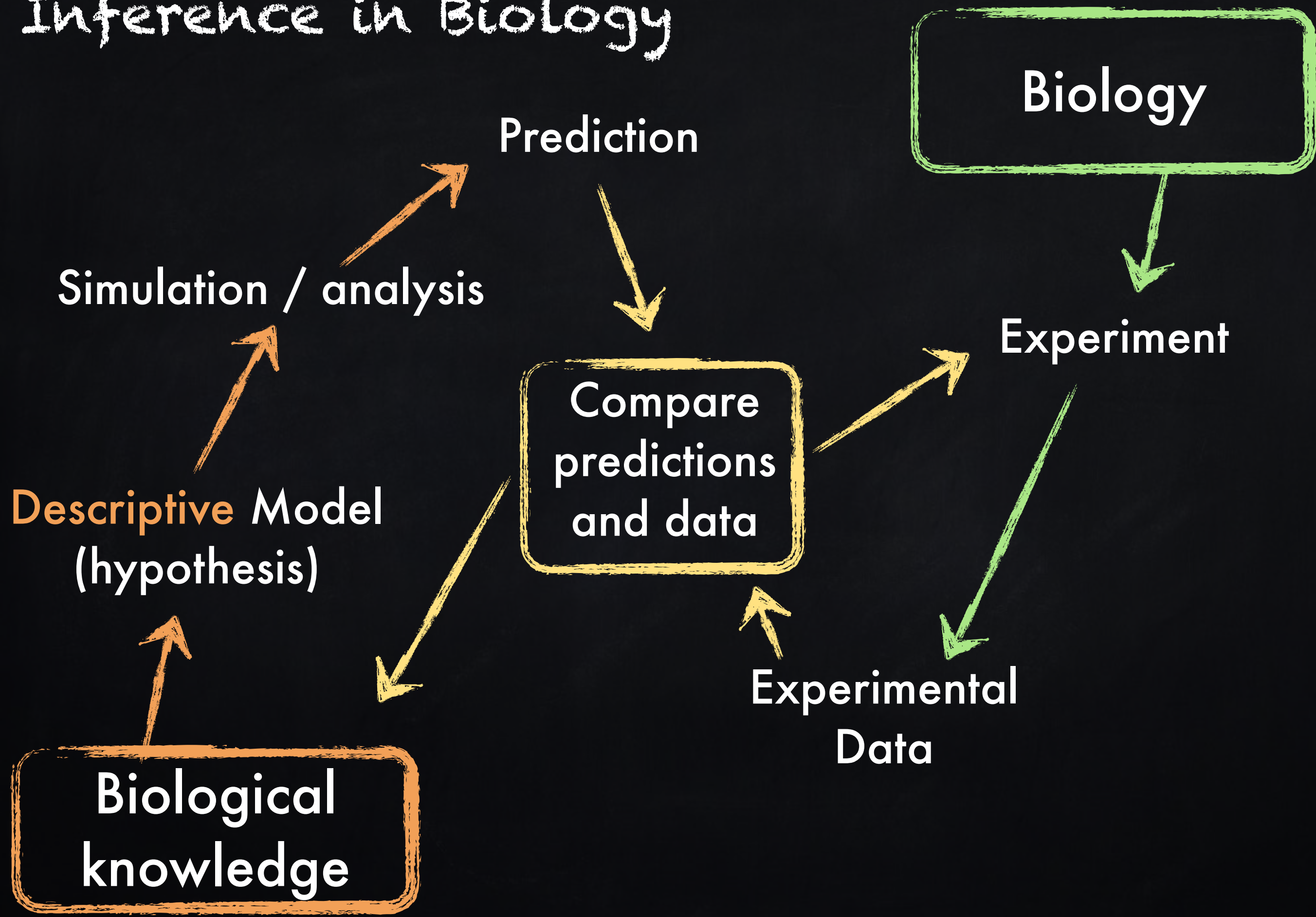
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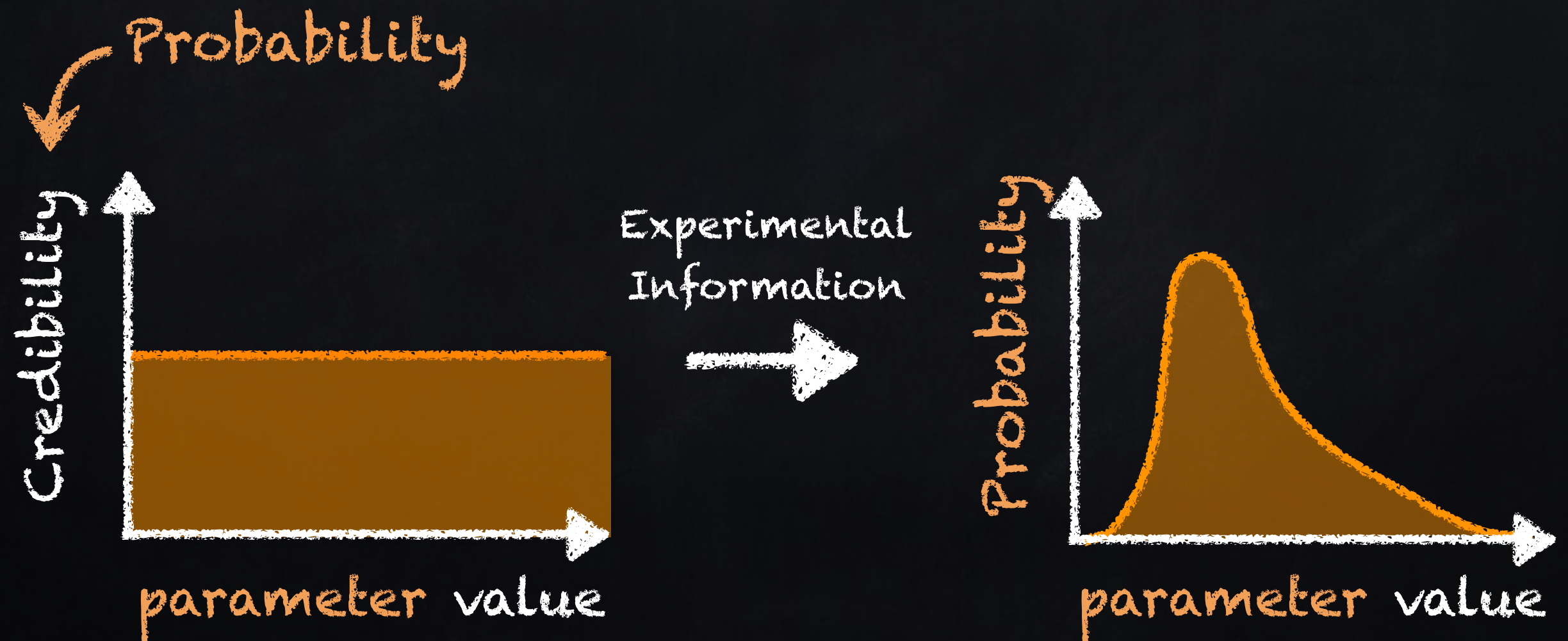
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Inference in Biology



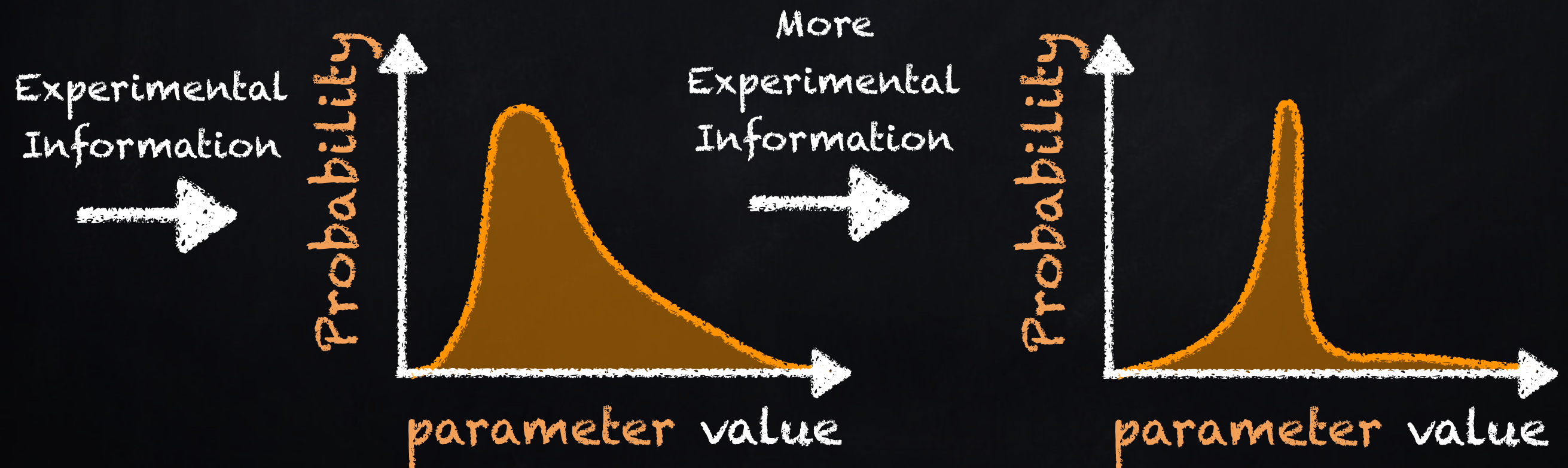
Bayesian inference is ...

... reallocation of credibility across possibilities
using **descriptive** mathematical models



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Probability 101

Random variable X : Possible outcome of an observation

Random variables can be discrete or continuous

Discrete Probability $P(X)$

impossible

determined

$$0 \leq P(X) \leq 1$$

normalisation

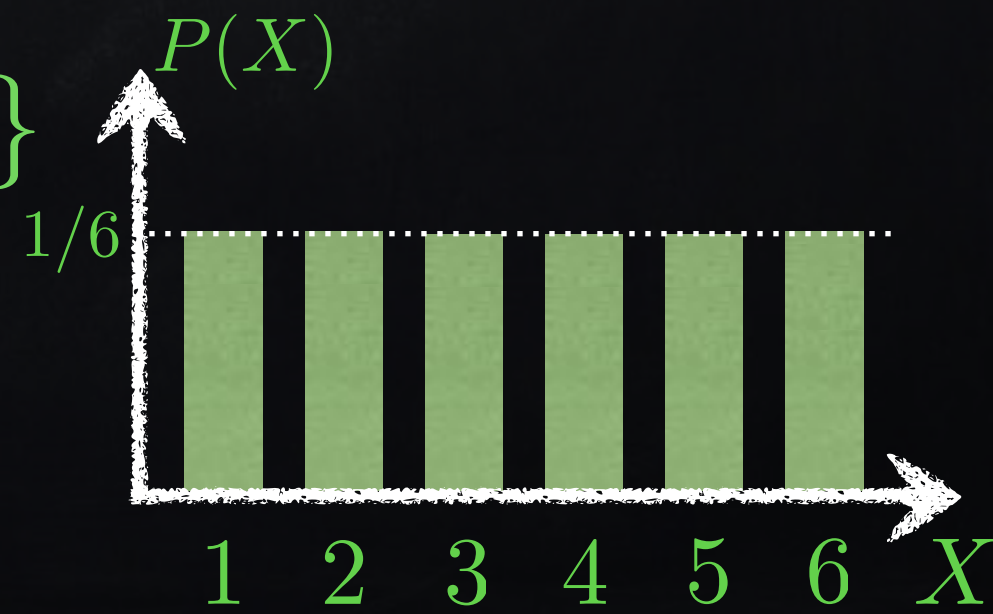
$$\sum_{\text{all outcomes}} P(X) = 1$$



$$X \in \{1, 2, 3, 4, 5, 6\}$$

X is discrete

$$P(X = 3) = \frac{1}{6}$$



Probability 101

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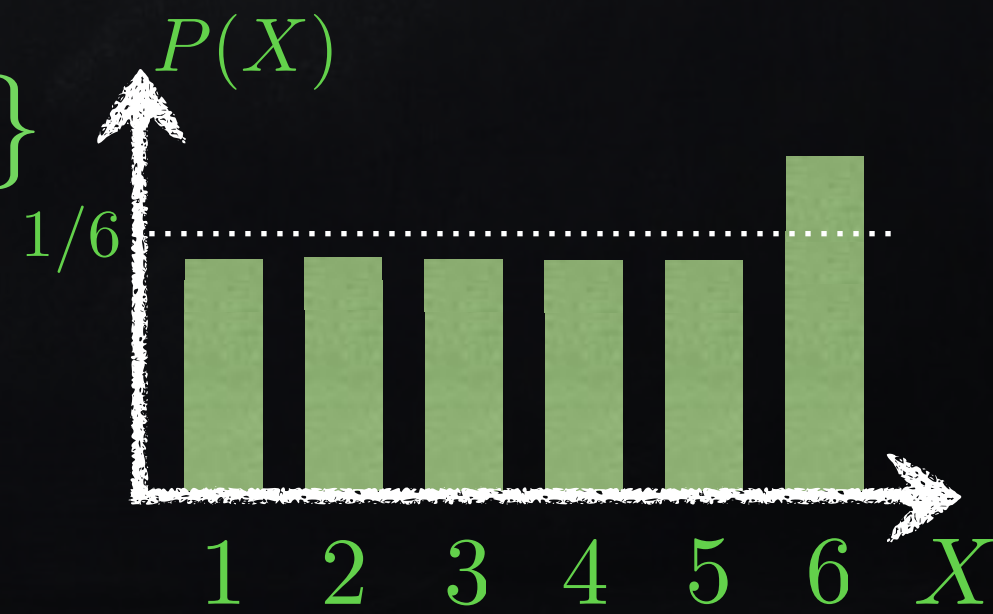
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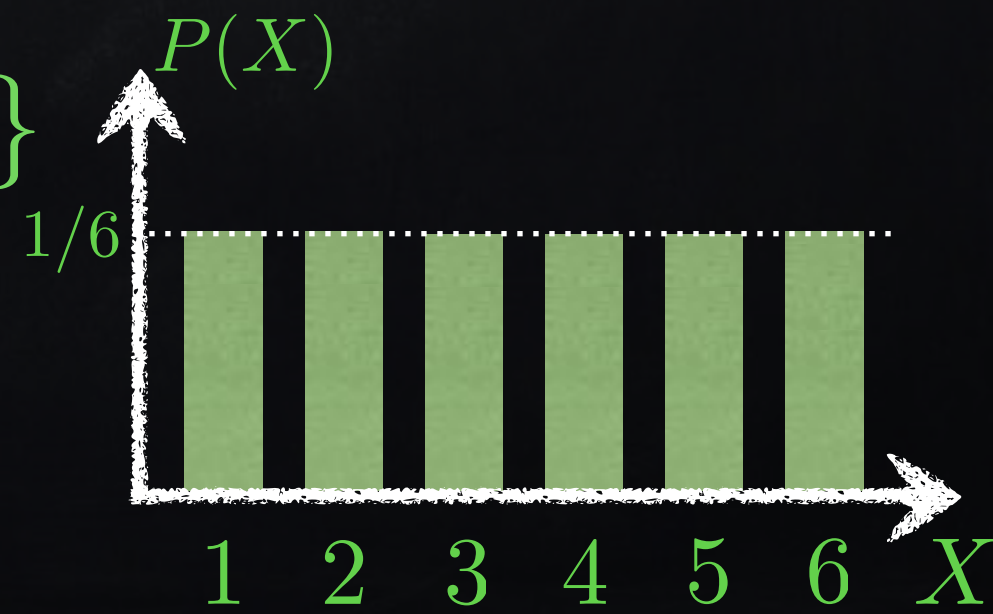
X is discrete

$$P(X = 3) = \frac{1}{6}$$

Common discrete distributions

- Uniform $P(X; N) = \frac{1}{N}$
- Poisson $P(X; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$
- Binomial

$$P(X; p, N) = \binom{N}{X} p^X (1 - p)^{N-X}$$



Probability 101

Random variable X : Possible outcomes

Random variables can be

Discrete Probability $P(X)$

impossible

deterministic

$$0 \leq P(X) \leq 1$$

normalisation

$$\sum_{\text{all outcomes}} P(X) = 1$$



$$X \in \{1, 2, \dots, 6\}$$

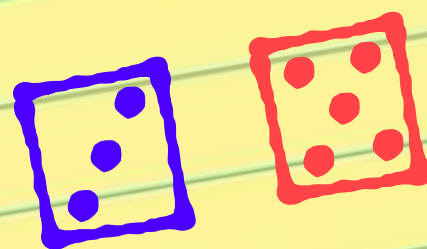
X is discrete

$$P(X = 3) = \frac{1}{6}$$

Question:

You have two regular 6-faced dice, one blue and one red.

What is the probability of throwing them and having the following result?



What is the probability of throwing both and getting a "3" in only one of them?
And at least one of them?

Probability 101

Random variable X : Possible outcome of an observation

Random variables can be discrete or continuous

Continuous probability $X \in \mathbb{R}$

$$\text{Probability}(A < X < B) = \int_A^B P(X) dX$$

$$\int_{\min X}^{\max X} P(X) dX = 1$$

normalisation

Probability density
function (PDF)

Common distributions

$$P(X, N) = \text{constant}$$

$$P(X; k) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2}$$

- Exponential

$$P(X; \lambda) = \lambda e^{-\lambda X}$$



$X \in [0, 80 \text{ km/h}]$
 X is continuous



Probability 101

Random variable X : Possible outcome of an observation

Random variables can be discrete or continuous

Continuous probability distributions

Probability($A < X < B$) =

$$\int_{\min X}^{\max X} P(X) dX = 1$$

normalisation



$X \in [a, b]$
 X is continuous

Question:

Find the value of A so the following distributions are PDFs

$$P(x) = A, \quad x \in [-1/2, 1/2]$$

$$P(x) = Ae^{-x}, \quad x \in [0, \infty]$$

X

Probability 101

2 random variables: X, Y

- Joint Probability $P(X, Y)$ Probability of an event from X AND an event from Y
- Conditional Probability $P(X|Y)$ Probability of an event from X GIVEN an event from Y
- Marginal Probability $P(X)$ Probability of an event from X INDEPENDENTLY from Y

$$P(X, Y) = P(X)P(Y|X)$$



$X = \text{month}$ $Y = \text{day number}$

$$P(\text{November}, 7) = 1/366$$

$$P(\text{November} | 7) = 1/12$$

$$P(7) = 12/366$$

Probability 101

2 random variables

Joint Probability

Conditional Probability

Marginal Probability

$$P(X, Y) = P(X)P(Y)$$



$X = \text{month}$

$P(\text{November})$

$P(\text{November})$

$P(7) = 1/12$

Question:

- calculate the following probabilities:

$P(31, \text{December})$

$P(31)$

$P(\text{Dec})$

$P(31 | \text{December})$

$P(\text{December} | 31)$

and test that:

$$P(31, \text{Dec}) = P(31 | \text{Dec})P(\text{Dec})$$

$$P(31, \text{Dec}) = P(\text{Dec} | 31)P(31)$$

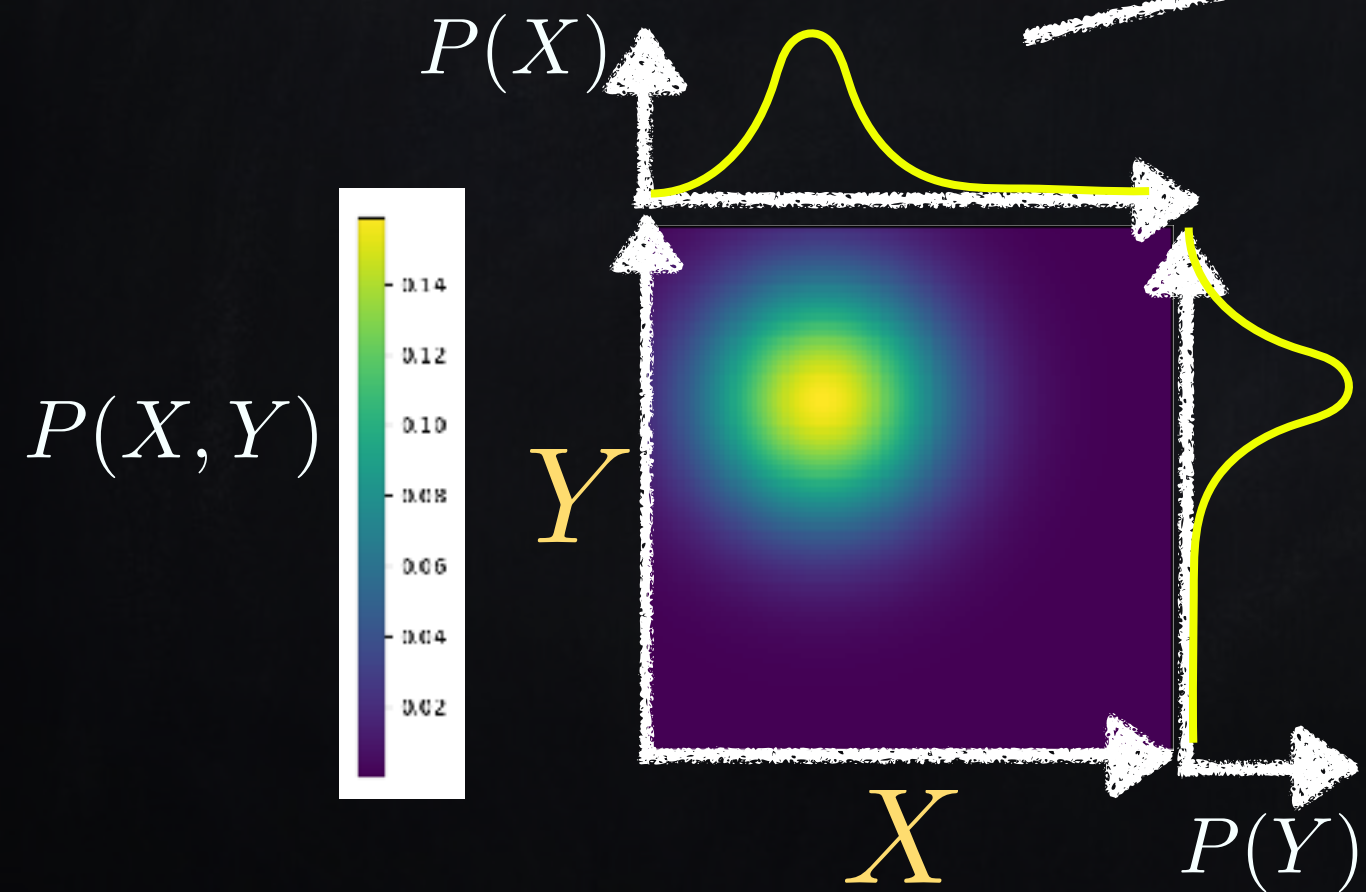
Probability 101

2 continuous random variables: X, Y

$$P(X, Y) = P(X)P(Y|X)$$

$$\int_{\min Y}^{\max Y} P(X, Y) dY$$

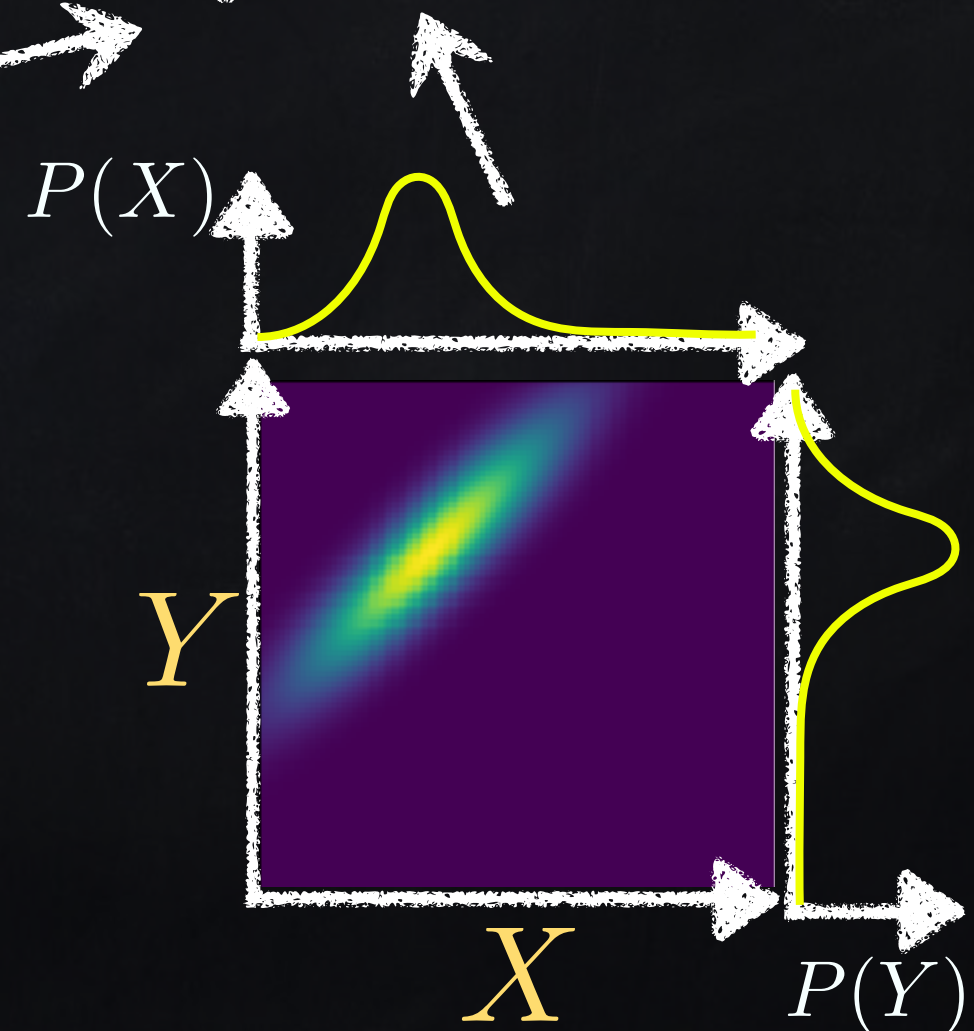
Same marginals but
different joint distributions!



Independent variables

$$P(Y|X) = P(Y)$$

$$P(X, Y) = P(X)P(Y)$$



Dependent variables

$$P(Y|X) \neq P(Y)$$

$$P(X, Y) \neq P(X)P(Y)$$

Probability 101

2 continuous random variables: X, Y

$$P(X, Y) = P(X)P(Y|X)$$

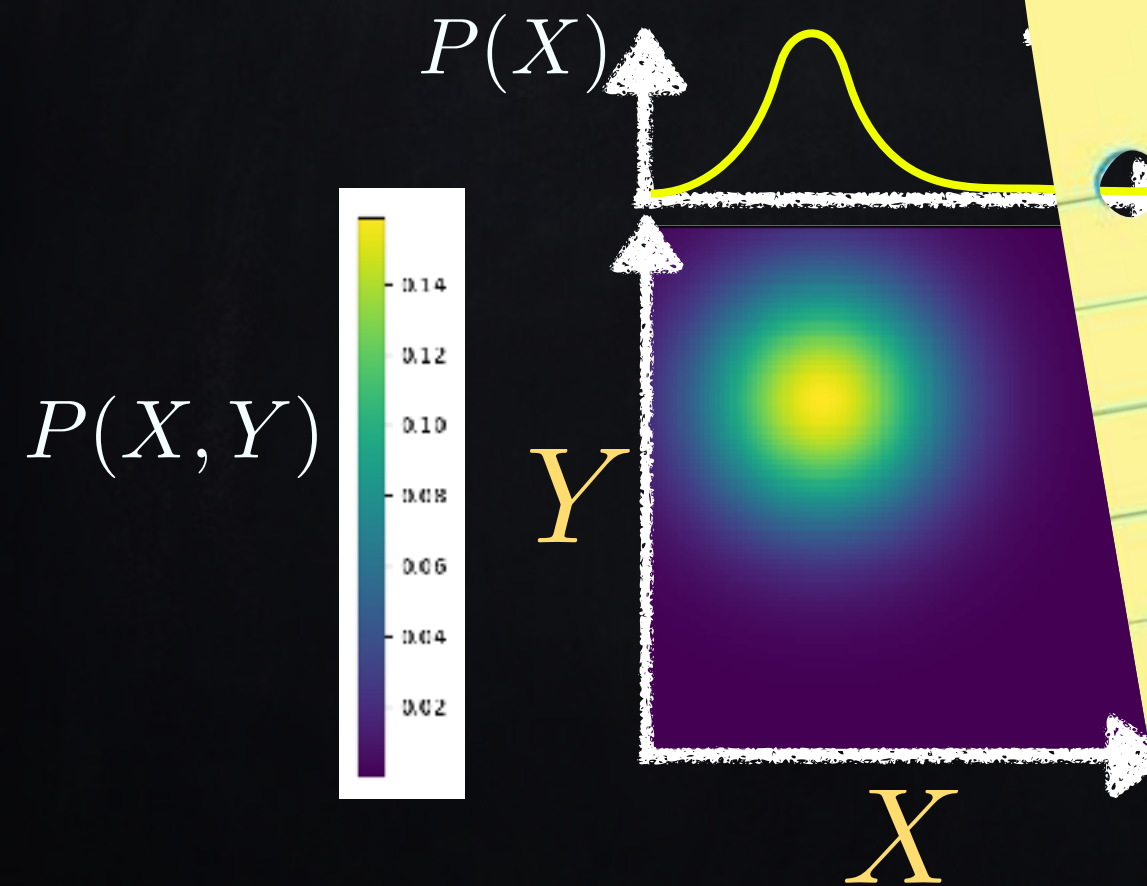
Same marginals but
different joint

$$\int_{\min Y}^{\max Y} P(X, Y) dY$$

Question:

- can we have a situation where X is a discrete random variable and Y a continuous one?

- what is the normalisation condition for $P(X, Y)$ where X and Y are continuous random variables?



Independent variables

$$P(Y|X) = P(Y)$$

$$P(X, Y) = P(X)P(Y)$$

Probability 101

Time t is usually
not a random variable



Probability in time $P(X, t)$ t can be discrete (e.g. generations)
or continuous (e.g. duration)

Steady state probability $P(X, t \rightarrow \infty) = \tilde{P}(X)$

t signifies sequential ordering i.e. **causality** in our model

Generative model:

$$P(X, t_N) = F(P(X, t_{N-1}), P(X, t_{N-2}), \dots, P(X, t_0))$$

Special case (Markov process)

$$P(X, t_N) = F(P(X, t_{N-1}))$$

e.g. Population dynamics,
Reaction kinetics
Epidemic models

Question:

In the simplest model of DNA substitution (JCG), each nucleotide can mutate to any other nucleotide with the same probability per unit time r .

Knowing that at a given time $t=0$ the base of site is A: $P(A, t=0) = 1$. Derive the general expression for the probability $P(G, t)$ of the site having a base G at time t .

1) Express the probability that the site has a base G at time $t + \Delta t$ as $P(G, t + \Delta t) = P(G, t) + F(t)$, where $F(t)$ represents the change in probability over the small time interval Δt .

2) Transform the above expression into a differential equation that describes the rate of change of $P(G, t)$ over time.

3) Solve the differential equation to find $P(G, t)$ expression for $P(G, t)$.

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Bayes' Theorem

$$P(X, Y) = P(X)P(Y|X)$$

$$P(X, Y) = P(Y)P(X|Y)$$

$$\frac{\cancel{P(X, Y)}}{\cancel{P(X, Y)}} = \frac{P(X)P(Y|X)}{P(Y)P(X|Y)}$$

$$P(X|Y) = \frac{P(X)P(Y|X)}{P(Y)}$$

Bayesian Inference

$$P(X|Y) = \frac{P(X)P(Y|X)}{P(Y)}$$



Prior probability

Likelihood

$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

Posterior probability

Y : Experimental data

X : Parameter value (δ)

Bayesian Inference

$$P(X|Y) = \frac{P(X)P(Y|X)}{P(Y)}$$

Likelihood

Prior
probability

$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

Posterior
probability

What about $P(data)$? ✓

How do I choose the prior? ✓

What am I supposed to do with the posterior? ✓

Okay, but how do apply this with my laptop? ✓

Can you give examples of the likelihood? ✓